

VLADIVOSTOK INTL, Russian Federation

WMO: 319741

Lat: 43.399N

Lon: 132.148E

Elev: 14

StdP: 101.16

Time Zone: 10.00 (E10)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-23.7	-31.1	0.2	-23.2	-29.1	0.3	-21.3	9.4	-10.9	8.6	-11.2	0.7	10	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	29.8	22.5	28.0	21.8	26.2	20.7	24.2	27.4	23.2	26.1	22.3	24.9	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	17.9	25.9	22.1	16.8	24.8	21.2	15.9	23.8	72.8	27.0	69.0	25.9	65.6	24.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.6	7.7	6.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-29.9	32.8	2.6	1.4	-31.7	33.8	-33.2	34.6	-34.7	35.4	-36.6	36.4
			-29.9	25.6	2.6	1.0	-31.8	26.3	-33.3	26.8	-34.7	27.4	-36.6	28.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-13.7	-10.2	-2.1	6.0	12.3	16.7	20.8	21.8	16.2	7.9	-2.1	-11.5	(6)
(7)		DBStd	12.93	4.33	4.99	4.87	3.29	2.94	2.69	2.70	2.62	3.32	4.22	5.79	4.85	(7)
(8)		HDD10.0	2938	735	566	374	127	12	0	0	0	2	92	364	667	(8)
(9)		HDD18.3	5001	993	799	632	371	189	63	8	4	79	323	614	925	(9)
(10)		CDD10.0	1204	0	0	0	6	82	200	336	366	187	27	0	0	(10)
(11)		CDD18.3	226	0	0	0	0	0	13	86	112	14	0	0	0	(11)
(12)		CDH23.3	1304	0	0	0	1	32	111	446	628	84	1	0	0	(12)
(13)		CDH26.7	292	0	0	0	0	2	22	118	143	7	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.8	3.0	3.2	3.3	3.2	2.9	2.6	2.4	2.4	2.7	2.7	2.8	(14)
(15)	Precipitation	PrecAvg	745	16	17	29	39	79	92	139	138	75	55	40	26	(15)
(16)		PrecMax	982	85	56	105	87	145	184	272	337	228	154	104	78	(16)
(17)		PrecMin	479	0	1	1	6	12	42	41	21	12	12	7	2	(17)
(18)		PrecStd	135	20	15	21	20	34	33	58	68	54	30	26	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	5.6	14.9	21.5	26.2	29.1	32.1	31.9	27.2	21.9	14.9	5.9	(19)
(20)		MCWB	-0.8	1.2	6.6	10.7	14.6	20.2	24.1	23.3	19.9	14.4	10.1	4.4	(20)	
(21)		2%	DB	-0.9	3.2	10.1	18.1	24.0	26.2	29.6	29.9	25.6	19.8	12.0	2.1	(21)
(22)		MCWB	-3.1	0.7	3.6	8.8	14.4	18.7	22.7	22.7	18.9	13.9	7.7	-0.4	(22)	
(23)		5%	DB	-3.0	1.8	7.9	15.8	21.8	24.0	27.9	28.1	24.0	18.0	9.1	-0.8	(23)
(24)		MCWB	-5.2	-0.5	2.5	7.7	13.2	18.1	22.0	22.1	18.1	12.6	5.8	-2.8	(24)	
(25)		10%	DB	-5.2	-0.5	5.6	13.1	19.1	21.9	25.9	26.8	22.2	16.1	6.8	-2.9	(25)
(26)		MCWB	-7.2	-2.9	1.2	6.4	12.4	17.3	21.3	21.5	17.1	11.8	4.2	-5.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	2.9	7.0	11.7	16.8	22.1	25.5	25.3	22.3	17.2	11.1	4.5	(27)
(28)		MDCB	1.0	4.0	13.0	19.8	23.9	27.8	29.7	28.7	24.2	19.4	13.4	5.3	(28)	
(29)		2%	WB	-3.2	1.3	4.6	9.9	15.3	20.1	24.1	24.4	20.7	15.6	8.7	0.1	(29)
(30)		MDCB	-1.5	2.8	8.8	15.8	21.2	24.1	28.1	27.5	23.2	18.3	11.1	1.2	(30)	
(31)		5%	WB	-5.5	-0.9	3.0	8.5	14.2	18.7	22.9	23.7	19.6	14.1	6.3	-2.5	(31)
(32)		MDCB	-3.3	1.3	6.7	13.4	19.5	22.5	25.9	26.4	22.6	17.1	8.4	-0.8	(32)	
(33)		10%	WB	-7.5	-3.1	1.7	7.2	13.2	17.7	21.9	22.9	18.5	12.5	3.9	-4.9	(33)
(34)		MDCB	-5.5	-0.7	4.8	11.6	17.9	21.0	24.6	25.3	21.5	15.3	6.3	-3.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.9	13.6	11.9	12.3	11.6	8.6	8.0	8.7	11.7	13.1	11.0	11.8	(35)	
(36)		5% DB	MCDBR	15.9	14.6	15.3	17.4	17.5	13.9	12.2	11.0	12.9	14.7	14.2	14.2	(36)
(37)		MCWBR	13.8	11.7	9.4	9.4	8.6	7.5	6.1	5.2	7.0	9.1	10.2	12.3	(37)	
(38)		5% WB	MCDBR	14.9	12.5	13.4	14.8	14.0	11.3	9.7	8.3	9.9	12.2	12.7	13.7	(38)
(39)		MCWBR	13.3	10.7	8.9	8.7	7.6	6.4	5.3	4.4	5.8	8.8	10.1	12.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.388	0.473	0.573	0.592	0.567	0.534	0.487	0.436	0.448	0.395	0.345	(40)	
(41)		taud	2.128	1.957	1.828	1.690	1.716	1.861	2.013	2.145	2.200	2.021	2.065	2.091	(41)	
(42)		Ebn at Noon	768	770	748	706	709	731	751	775	785	699	673	698	(42)	
(43)		Edh at Noon	100	142	184	229	230	199	170	144	126	133	107	95	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.21	3.18	4.20	4.64	5.02	5.11	4.59	4.50	4.20	3.12	2.17	1.81	(44)	
(45)		RadStd	0.13	0.19	0.27	0.43	0.45	0.67	0.60	0.44	0.37	0.24	0.18	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page